Status	Finished
Started	Wednesday, 21 October 2026, 11:18 AM
Completed	Wednesday, 21 October 2026, 11:27 AM
Duration	9 mins 6 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It can only be applied to large lists
c. It is slower than linear search for small lists
d. It requires the list to be sorted ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(1)
c. O(n log n)
d. O👎 ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Linear search ✓
c. Hash search
d. Both (1) & (3)

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The search stops

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By using a hash function

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. log n
c. n/2
d. 1

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 10 ✓
c. 500
d. 1000

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the left sublist ✓
c. At the beginning of the list
d. In the right sublist

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list is sorted
c. When the list contains duplicate elements
d. When the element is found in the list ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. The first element is returned
c. The last element is returned
d. An error is raised

Question **11**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Mining
c. Searching ✓
d. Finding

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By using a hash function
c. By starting from the first element
d. By dividing the list length by two ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O(1)
c. O(n log n)
d. O👎

Question **15**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search continues until the end of the list
d. The search backtracks to find duplicate elements